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Introduction

This paper is divided into two parts according to two assignments.

- Assignment 1 : Interplanetary transfer. The objective is to design a transfer from Jupiter to Asteroid N.137052 with a flyby at Mars. The final result should be to present a possible transfer with minimum cost using patched conics approach. The mission cost is assumed to be only dependant on the total ΔV . The mission begins with the satellite having the same velocity as Jupiter. Then it requires an initial propulsion manoeuvre to input into the first leg of the transfer. A coasting phase follows until the satellite reaches the Mars' sphere of influence. Here a powered flyby takes place. Now on the second transfer leg, after another coasting phase, the satellite intercepts the asteroid and conducts the final burn to input into the asteroid's orbit.

Assumptions :

- Impulsive manoeuvres.
 - Mars' sphere of influence is neglected in the heliocentric frame.
 - Transfer orbits found by 2 body problem solutions.
 - No collisions with other space objects during the whole mission.
 - Instantaneous change of velocity at position/time of flyby in heliocentric frame.
 - Powered gravity assist burn conducted at the pericentre of the hyperbolas.
- Assignment 2 : Planetary explorer mission on Mars. The goal is to carry out the orbit analysis and ground track estimation of an orbiter deployed around Mars during the fly-by of the interplanetary mission. To achieve this, the J2 and Solar Radiation Pressure (SRP) perturbations are modelled and the different results are discussed. In the propagation of the elements, two approaches have been considered: using Kepler equations in Cartesian coordinates and using Gauss Planetary equations. Finally, an analysis of the evolution of the elements of two real uncontrolled objects orbiting Earth has been carried out.

Assignment 1 : Interplanetary Transfer

1.1 Preliminary Analysis

The overall mission time frame, from the earliest departure from Jupiter and the latest arrival on the asteroid are displayed in table 1a. Due to the computational cost involved in finding an optimal interplanetary transfer over the full

First departure	2030/01/01 at 00:00:00
Latest arrival	2060/01/01 at 00:00:00

(a) Laucher assigned times

Departure window	2030/01/01	2042/01/01
Flyby window	2030/09/01	2046/01/01
Arrival window	2030/09/01	2048/01/01

(b) Reduced time windows

time frame, a refinement of the search window was made. In the absence of a 3-body synodic period, the synodic periods were calculated two-body pairs at a time. First, the synodic period between Jupiter and Mars, $T_{M-J} = 2.24$ years and the synodic period between Mars and the asteroid, $T_{M-A} = 5.39$ years using the very rough assumption of circular coplanar orbits. Expanding upon these results, the two results are multiplied by integers to determine when the solutions overlap. Applying this method a time of 4082 days is found, a little more than 11 years yield a close repetition. To adress the inaccuracy caused by the initial assumption, the time window was further increased to 12 years as a margin. It is then assumed that any solution found will approximately repeat every 12 years.

A second constraint is introduced with the time of flight of the first leg, from Jupiter to Mars. The minimum time needed to go from Jupiter to Mars was calculated using the parabolic transfer. The parabola is going from the Jupiter pericentre distance to the Mars apocentre distance. This approach is feasible thanks to the two objects having almost coplanar orbits. $\Delta t_{min} = 243$ days was computed. The maximum time of flight of leg 1 could be calculated considering the TOF on a Hohmann type transfer when the planets are the furthest, 1227 days. To include non-Hohmann transfers this estimate is further increased to 4 years. A similar reasoning is made on the second leg of the transfer, from Mars to the Asteroid. Noting that the two orbits have a significant difference in inclination and the asteroid's orbit having high eccentricity, and the fact the orbits pass really close to each other, no assumption is made on the minimum time of flight for leg 2. For the maximum TOF of leg 2 firstly the period of an ellipse connecting coplanar apocentres of the two orbits is computed, 502 days. To be even more cautious, 2 years is chosen as the maximum TOF. These windows are shown in table 1b. Finally, the flyby is considered to be doable at a minimum altitude of 11 km from the surface of Mars, due to the presence of its atmosphere.

1.2 Pork-chop plots

A lot of insight in the design of the search window can be gained by dividing the total cost of the mission into the cost of the different legs. In figure 1 the pork-chop plot of the departure burn can be seen with the departure date on the horizontal axis and the time of flight on the vertical.

Similiar plots can be made summing the Δv cost of the flyby and the rendez-vous with the asteroid. However, due to the fact that the cost of the flyby depends on the connecting leg means that several pork chop plots are necessary to show the effect of the time of flight of the first leg. The maximum turn angle achievable in the flyby is limited meaning only a small subset of the transfers in leg 1 and leg 2 that are possible to connect. However, even flybys that aren't possible may give insight and so are also shown, an overlay in red is added to highlight the points where a flyby is possible. Looking at figure 1, the cheapest transfer corresponds to a flight time of around 1200 days regardless of the departure date. Discontinuousness are found periodically corresponding to half the synodic period between Jupiter and Mars. They are caused by singularities in the Lambert algorithm when the transfer is 180 deg or 360 deg. Comparing common points between the plots in figure 2 it can be seen that varying the flight time of the first leg from 600 to 1500 days does not significantly change the ΔV required.

This means that finding a possible interplanetary trajec-

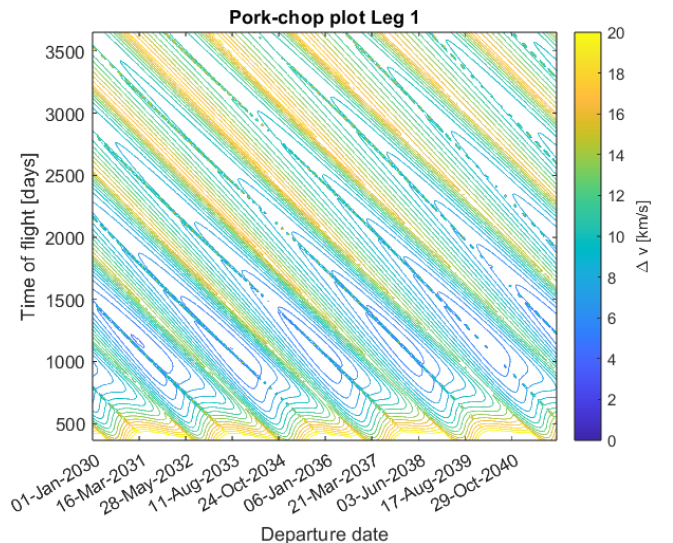


Figure 1: Pork chop plot showing the departure burn required ΔV for the first interplanetary leg.

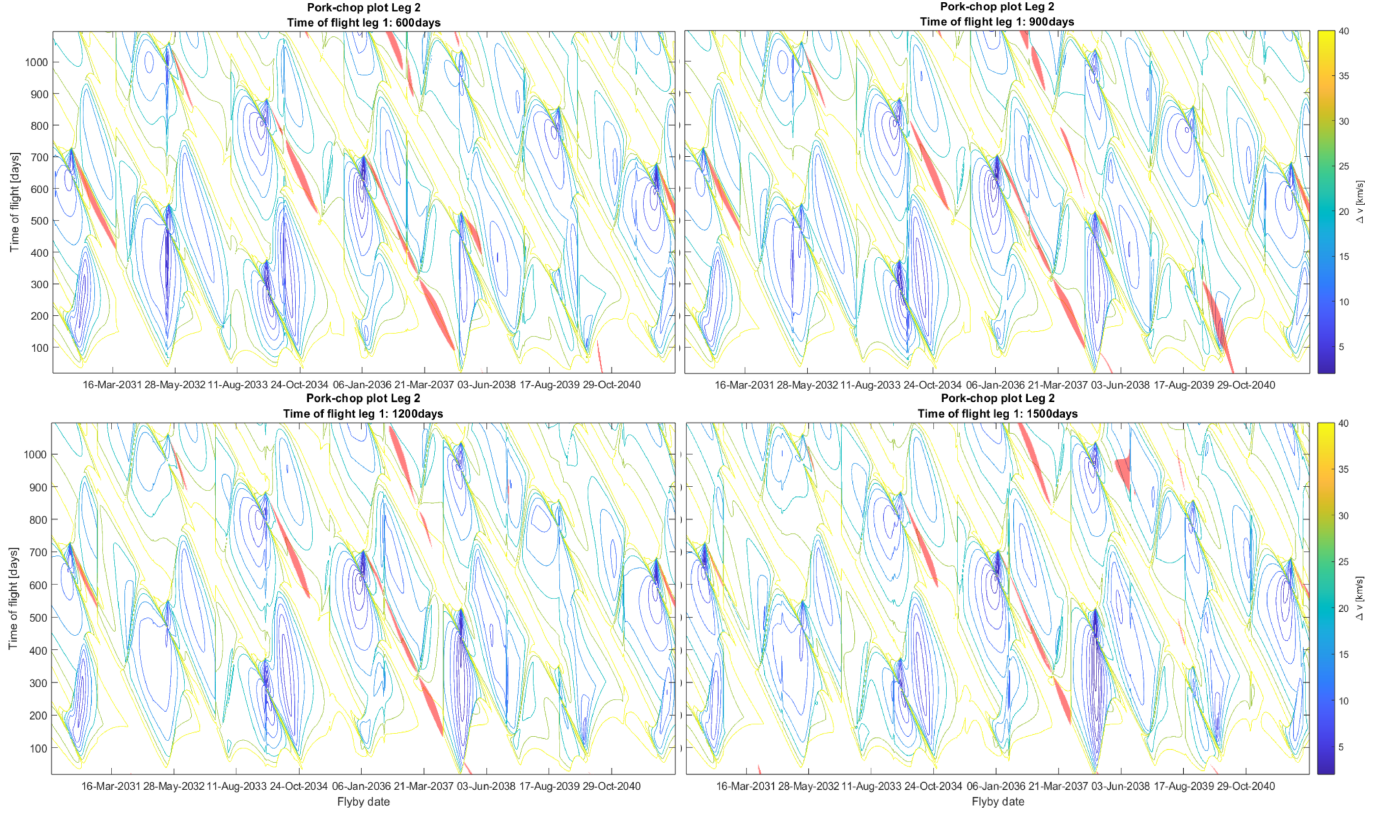


Figure 2: Pork chop plots showing the arrival rendezvous manoeuvre and flyby burn required ΔV in connection to the first leg through different time of flights. Areas in red show regions where a flyby is physically possible.

tory can be done graphically by finding the flyby date needing a low amount of ΔV while being located in one of the red zones.

1.3 Grid search

The departure, flyby and arrival dates means a nested triple *for* loop approach was used to find a suitable minimum ΔV . From which a time vector could be fed into Matlab's *fmincon* function in order to find the local/global minima corresponding to the initial condition found through the *for* loop. The nested loop was built as follows :

- The external loop iterates for departure dates. Calculates Keplerian and Cartesian elements of Jupiter at each step.
- The second loop iterates for flyby dates and calculates the Keplerian and Cartesian elements of Mars at each step. These values are then used in *Lambert.m*. The Lambert algorithm is used to find zero revolution, prograde, transfers from one point in space to another that satisfies the first leg's time of flight. Then the cost for the first burn done at departure from Jupiter is computed following the output departure velocity of the Lambert arc.
- The third loop iterates for arrival dates and calculates the Keplerian and Cartesian elements of the asteroid at each step. They are then used to find the second leg using again the Lambert's algorithm. Then the last burn at the asteroid is found. Finally, the velocities from leg one arrival at Mars and velocity of leg 2 leaving Mars are used to compute, if possible, the cost for the burn at the pericentre of the flyby. If the flyby is not possible the whole iteration is stored with a total $\Delta V = NaN$.

When the *for* loops gives finite ΔV the total value is calculated as the sum of the three separate burns. Then the minimum in this matrix is found, and the corresponding departure, flyby and arrival times are located. These are then fed into *fmincon* as the initial point. The three-dimensional matrix containing ΔV is heavily dependant on the step size of each iteration, for some trajectories, even though feasible in a specific hour of a date, are in some cases not at all feasible at a distance of a couple of hours in proximity. Once a local minimum is found it is reasonable to restart the loop using a different step size before feeding into *fmincon*. For this process, the time windows are refined and stepsize is decreased.

1.4 Results

The initial nested loop was run with a time step of 100 days for all three windows. It was able to find a minimum of $\Delta V = 22.5332$. Re-iterating the nested loop, in a two-year window around this result with, respectively 60,30,30 days time steps, a total $\Delta V = 16.5096$ is computed. Finally, starting with this result as the initial condition, the *fmincon* function found a result of $\Delta V = 15.8850$ km/s. In tables 2a,2b data regarding this final result are displayed:

Departure	2037/07/31 at 13:00:59
Flyby	2040/04/26 at 14:12:49
Arrival	2040/09/06 at 03:30:54

(a) Optimal departure, flyby and arrival dates

Time of flight leg one	1000.05 days
Time of flight leg two	135.55 days

(b) Optimal time of flights.

The complete heliocentric trajectory is shown in figure 3 and characterized as shown in table 3.

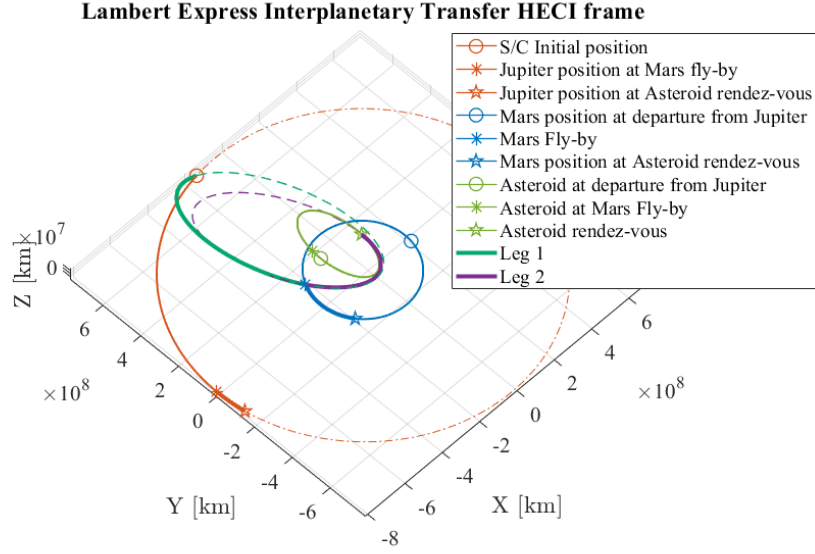


Figure 3: Interplanetary trajectory plot.

Time	a (km)	e (—)	i (deg)	Ω (deg)	ω (deg)	θ (deg)	ΔV (km/s)
Departure	7.78×10^8	0.0486	1.3009	100.8343	274.1036	85.9865	8.8020
Departure	4.28×10^8	0.8751	2.2567	100.8725	185.8835	174.1684	-
Flyby	4.28×10^8	0.8751	2.2567	100.8725	185.8835	226.8810	gravity assist
Flyby	3.78×10^8	0.8759	3.0912	118.1107	173.1302	222.4142	-
Arrival	3.78×10^8	0.8759	3.0912	118.1107	173.1302	122.4629	7.0829
Arrival	1.87×10^8	0.8095	14.9071	64.2756	209.9109	139.1914	-

Table 3: Orbit characterization and manoeuvring points.

The flyby manoeuvre is characterized by the results shown in table 4 :

	rp (km)	v_∞ (km/s)	v_p (km/s)	a (km)	e (—)	i (deg)	Ω (deg)	ω (deg)	θ_{SOI} (deg)
Entry	3401	24.5139	25.0223	-71.2699	48.7202	15.1667	352.9238	269.2463	-90.8315
Exit	3401	24.5139	25.0223	-71.2700	48.7201	15.1667	352.9238	269.2463	90.8315

Height (km)	δ (deg)	Δv_{ga} (km/s)	ΔV_{fb} (km/s)	$\Delta v_{ga}/\Delta V_{fb}$	Flyby time (h)
11.11	2.35	3.1700×10^{-5}	1.0063	0.000 032	13.07

Table 4: Flyby.

The flyby time is calculated considering the finite sphere of influence, $r_{SOI} = 5.77 \times 10^5$ km. This means that the true anomaly at entry and exit are not exactly those given by the hyperbola asymptotes. As seen in the result table 4, the hyperbolas have a high eccentricity and the total turn angle is also small. Although this seems to not change much in the Mars-centric transfer, in the heliocentric frame it is appreciable how the second leg has a completely different orbit due to the powered gravity assist. The flyby in the Mars-centric inertial frame can be seen in figure 4.

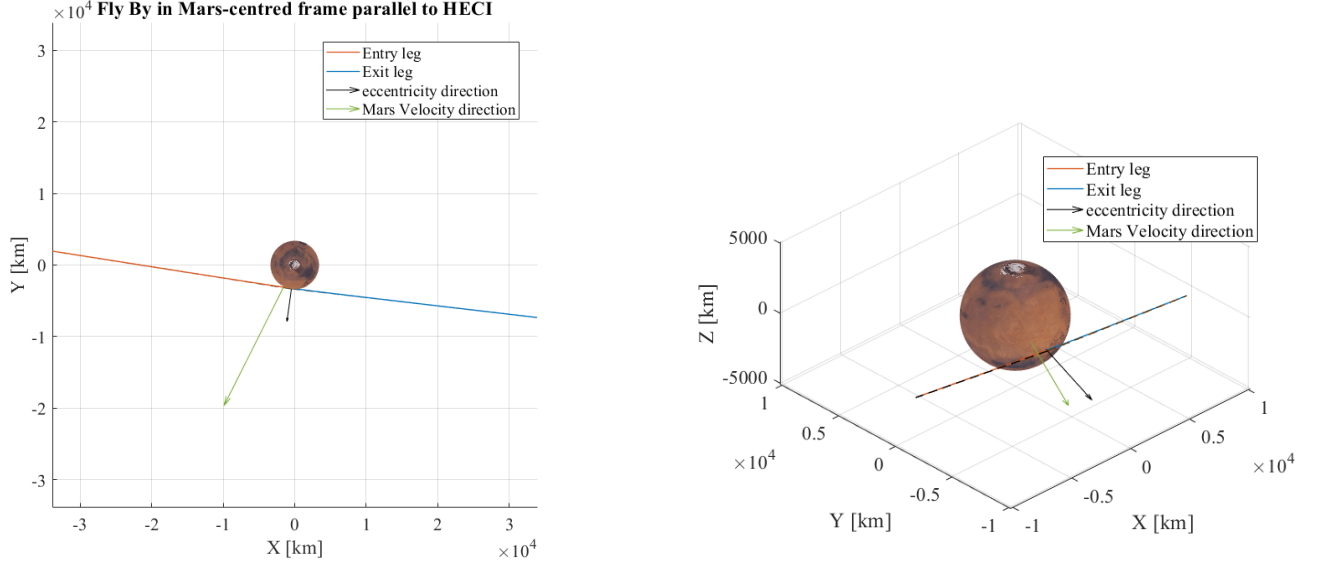


Figure 4: Mars Flyby

1.5 Discussion

The results found for the interplanetary transfer trajectory align well with the pork shop plots shown in section 1.2. Some differences are expected since the resolution of the search in the optimization is higher.

The first leg leaves in the vicinity of an acceptable flyby in figure 1 and corresponds to a TOF of 1000 days. That is why the fly by, corresponding to a time of flight of leg 2 of 135.55 days is located in figure 2 on the y axis in an acceptable flyby region. While flyby date, due to the dependency on the TOF of leg 1 is accordingly on a shifted x-axis location. Another noticeable fact is that the flyby occurs when Mars is near its apocentre. Multiple iterations with different time windows showed that the local minimum is always found when the powered gravity assist is carried out near Mars orbit's apocentre. Also Jupiter and the asteroid, in the minimum cost solutions, are in the same regions of their orbits. This behaviour is particularly clear when looking at a difference in time of the defined synodic period from the found solution. It shows that the preliminarily defined synodic period practically applicable. Furthermore, looking at the inclinations of the orbits it could be inferred that the most costly burn is going to be the one needed to change inclination. This change is incrementally done through the three burns. A slight change in the first burn, then another small change during the flyby, though the most appreciable change in inclination is done at rendez-vous with the asteroid. Thanks to this step by step inclination change and due to the fact the burns are conducted not only to change the inclination, the departure burn remains still the largest one. Finally, the needed gravity assist burn is so small compared to the total velocity change achieved from the flyby that it seems almost a natural flyby. The assumption of burn conducted at the pericentre to make such a precise adjustment is unthinkable in real life, where the burn would be conducted with continuous burns through out the hyperbola.

Assignment 2 : Planetary Explorer

2.1 Definition of the orbit and perturbations

The orbit considered for the study is partially defined by the mission requirements. While initial values for the semi-major axis, the eccentricity and the inclination were imposed, the rest of the elements were chosen attending at their effect on the ground track. While different values of Ω and θ only causes a longitudinal shift of the ground track (positive angles move eastward, negative angles move westward), the value of ω has a notable effect on the shape of the orbit (and therefore also in the ground track). $\omega = 90$ deg is a value that provides East-West symmetry and a more regular shape, favourable for potential study missions of Mars' equatorial surface.

Element	a (km)	e (-)	i (deg)	Ω (deg)	ω (deg)	θ (deg)
Value	21400	0.0781	3.47	10	90	0

Table 5: Orbit initial elements.

It is relevant to point out that the semimajor axis value is above the aerostationary one (20.428 km), so we can expect that the evolution in time of the ground track will be in the West direction. The initial time, when the orbiter explorer is deployed, is set to 2040-04-26, aligned with the time of the fly-by from the first assignment. The initial shape of the orbit is shown in Figure 5.

The perturbations assigned are J2 and Solar Radiation Pressure (SRP). The J2 effect is a result of the Earth oblateness, which creates oscillations in every element at different time scales, as well as secular variations in Ω , ω and θ . Due to the value of the inclination, westward regression of the nodes and positive periapsis precession are expected in the results. On the other hand, SRP is a perturbation caused by the impact of photons emitted by the Sun. For Mars, the solar irradiance is equal to $I = 586.2 \text{ W/m}^2$. The SRP is greatly influenced by the reflectivity of the surface of the satellite and its area over mass ratio, which have been imposed as $c_R = 1$ and $A/m = 1 \text{ m}^2/\text{kg}$, respectively.

2.2 Groundtrack analysis

In this Section, an analysis of different ground tracks is carried out to understand the area of Mars covered during the explorer mission. For the analysis, both unperturbed and perturbed evolutions are shown and three different propagation times have been considered:

- **1 period:** this time interval, comparable to an Earth day (about 1.1 Earth days), allows us to see the shape of the nominal orbit, and the coordinates (longitude and latitude) covered after one revolution around Mars since the arrival. The effect of perturbations is not visible at this point (Figure 6).

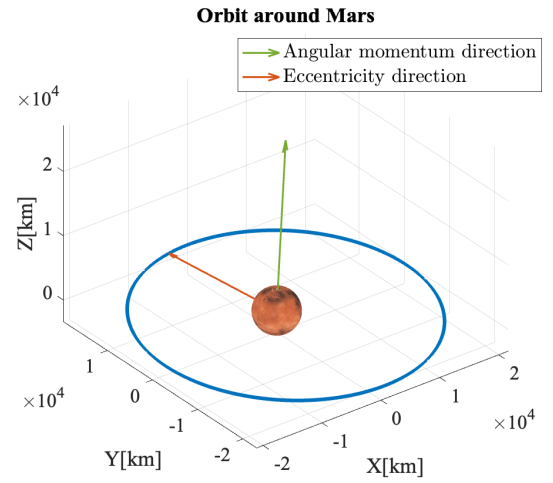


Figure 5: Initial shape of nominal orbit.

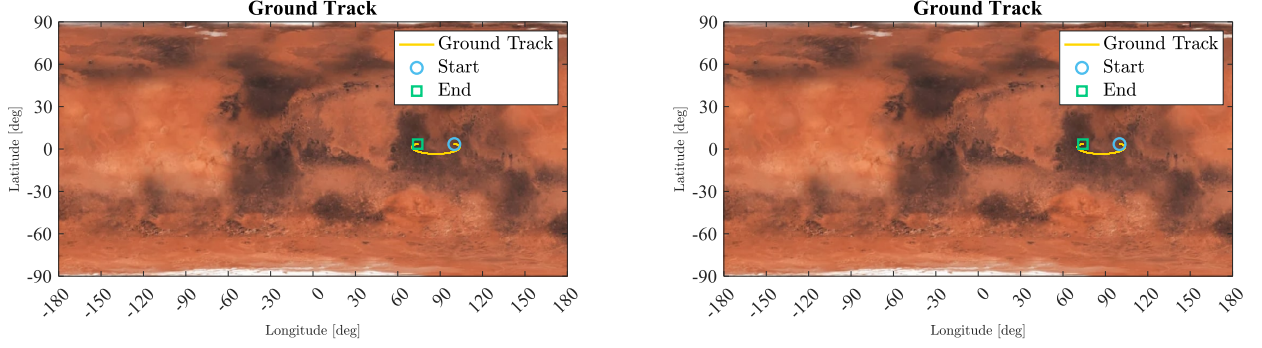


Figure 6: Unperturbed and perturbed nominal ground track for 1 period.

- **13.618 periods:** represents the amount of time necessary to cover the entire perimeter around Mars, that is, how much it takes to reach condition $\lambda(t_f) = \lambda(t_0)$. In this case, a slight deviation between the initial and final point can be appreciated (Figure 7).

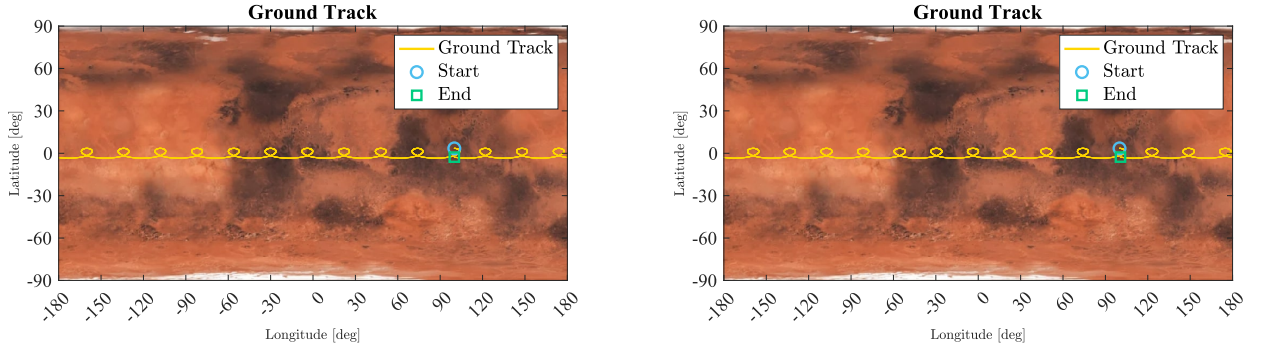


Figure 7: Unperturbed and perturbed nominal ground track for 13.618 periods.

- **2 months:** a sufficient time to better visualize the effect of the perturbations. It can be observed that the ground track, as time progresses, tends to shift eastward. This causes the final point of the perturbed ground track to be at -14 degrees, while the unperturbed one is at -20 degrees. This cumulative 6 degrees eastward shift (which would be even greater if we simulated the orbit for a longer time) is obviously caused by the J2 and SRP disturbances. The main contribution comes from J2, and this eastward shift is related to the fact that, with an inclination lower than 63.4 degrees, there is a positive precession of the argument of pericentre, which is reflected in this behaviour of the ground track. (Figure 8).

Certain missions, such as Earth observation and weather forecasting, rely on regularly covering the same areas to capture images or data under consistent lighting and viewing conditions or to monitor atmospheric patterns. To meet these requirements, repeating ground tracks are often essential. In order to replicate this idea on Mars, both the number of revolutions of the satellite ($k = 4$) and the rotations of the planet ($m = 1$) have an influence on the repeating ground track. These parameters provide a new value for the semi-major axis as:

$$n = \omega_M \frac{k}{m} \longrightarrow a_{rep} = \sqrt[3]{\frac{\mu_M}{n^2}} = 8106.71 \text{ km} \quad (1)$$

The result of a 4 period simulation is shown in Figure 9, for both the unperturbed and perturbed evolutions of the repeating ground track.

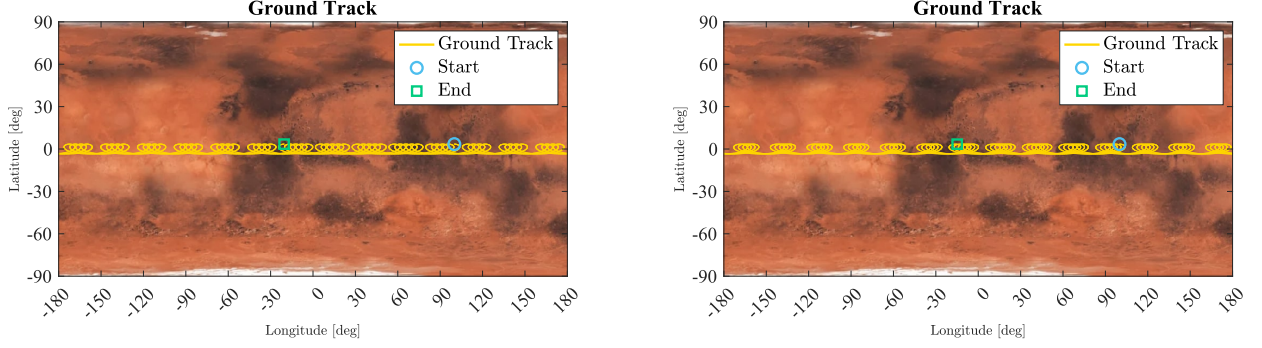


Figure 8: Unperturbed and perturbed nominal ground track for 2 months.

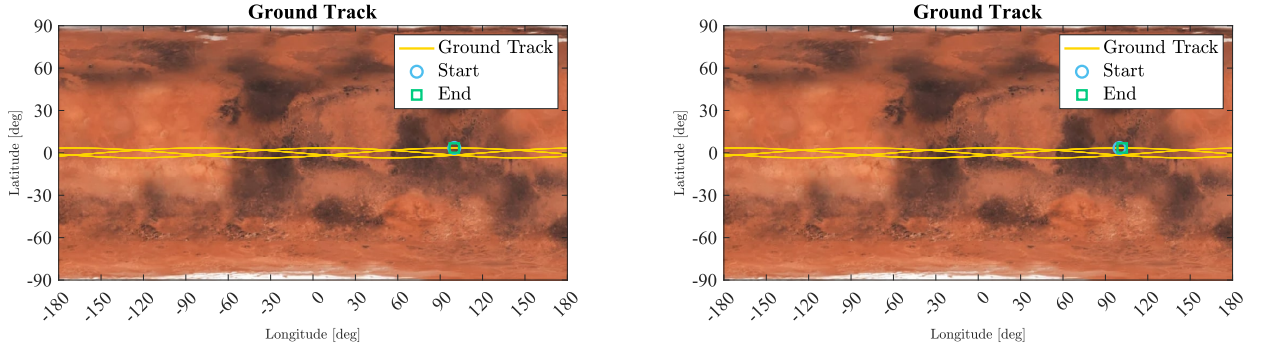


Figure 9: Unperturbed and perturbed repeating ground track.

Even in the repeating ground track, it can be observed how the presence of disturbances slightly changes the shape and causes an inaccurate repetition after 4 revolutions around Mars. In particular we can notice also in this case a shifting towards east of the final point, again caused by the effect of J2 and SRP, as discussed before. It can be predicted that, by analysing an increasing number of periods, the repetition will become more degraded, proving that in order to have a repeated ground track in real life, it is necessary to actively control the S/C.

2.3 Evolution of the orbit

In order to be able to recognize all the effects of the assigned perturbations, the propagation time has been set to 750 periods, which equals an amount of time slightly over a Martian year. The propagation has been carried out in Cartesian formulation using Kepler equation, along with a Keplerian formulation with Gauss Planetary equations. Figure 10 presents two plots illustrating the orbital evolution over time. The first plot, on the left, uses uniform scales across all three dimensions, making it challenging to clearly observe the orbital changes. In contrast, the second plot, on the right, employs different scales for each dimension, allowing for a clearer visualization of variations in inclination, argument of periapsis, and right ascension of the ascending node.

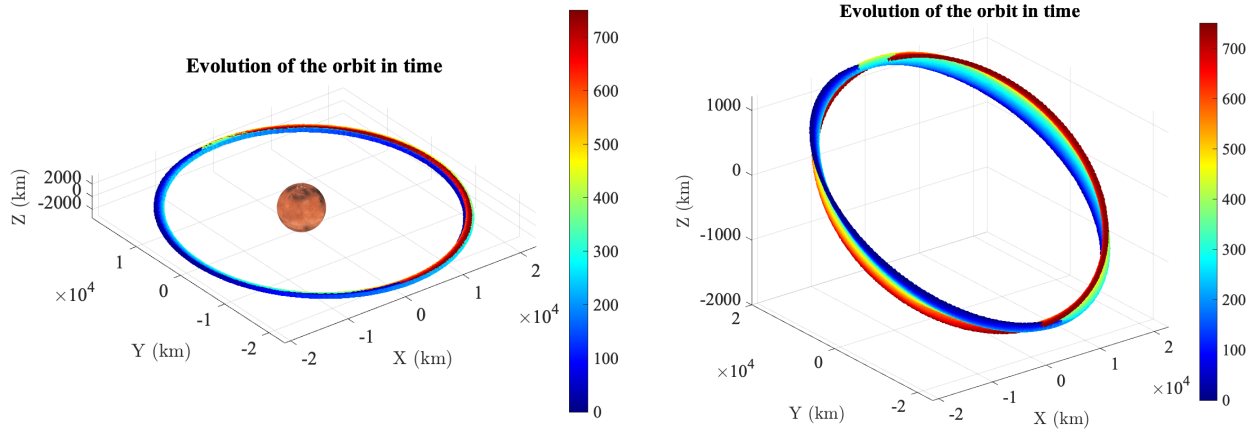


Figure 10: Evolution of the orbit.

2.4 Evolution of Keplerian elements

In this Section, an analysis of the effects of the assigned perturbations on the Keplerian elements is carried out, comparing both Cartesian and Gauss formulations. Therefore, the explorer is simultaneously subject to the effects of J2 and SRP. Both of them cause short-term variations of approximately one orbit period and secular variations, and the SRP also causes long-term variations of approximately one year period. Eventually, in the filtering part will be discovered that there are more long-term variations with longer periods than the ones resulting from SRP. Regarding the different elements:

- **Semimajor axis:** short-term variations can be seen due to both J2 and SRP, being the second one the most relevant one. There is no apparent secular evolution.
- **Eccentricity and inclination:** subject to short-term variations due mainly to J2 and slightly to SRP, and long-term variation exclusively due to SRP. Again, there is no apparent secular evolution.
- **RAAN and argument of periapsis:** they have a short-period variation due to both perturbations, a long-period variation due to SRP, and a secular variation given by the combination of the disturbances, but being J2 the most relevant one. The secular evolution is negative for Ω (nodal regression) and positive for w (perigee precession), consistent with the theory, which links their behavior with the value of the inclination: $i < 63.4^\circ < 90^\circ$
- **True anomaly:** it presents a short-period variation, as well as a secular evolution as a result from the effect of J2.

In Figure 11, it can be appreciated that the results obtained from both methods are apparently the same, so the relative errors for each element, plotted in semi-logarithmic scales, are really small. The relative errors are shown in Figure 12. Moreover, reference computational times for a propagation time of 750 periods and 50.000 intervals can be seen in Table 6, being the Gaussian propagation more convenient.

	Cartesian	Gaussian
Computational time [s]	2.814	1.634

Table 6: Computational time comparison.

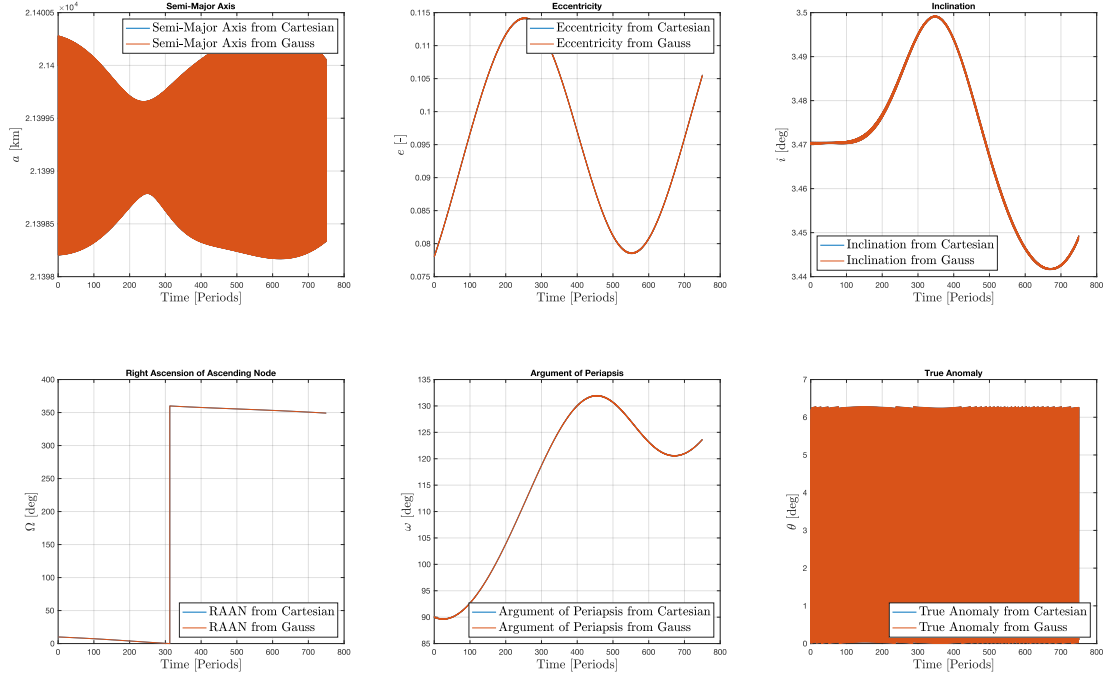


Figure 11: Evolution of Keplerian elements.

2.5 Filtering of Keplerian elements

From the evolution of the Keplerian elements, different frequency oscillations have been characterized. These oscillations can be removed applying a low-pass filter, then retrieving the longer-period or secular evolutions.

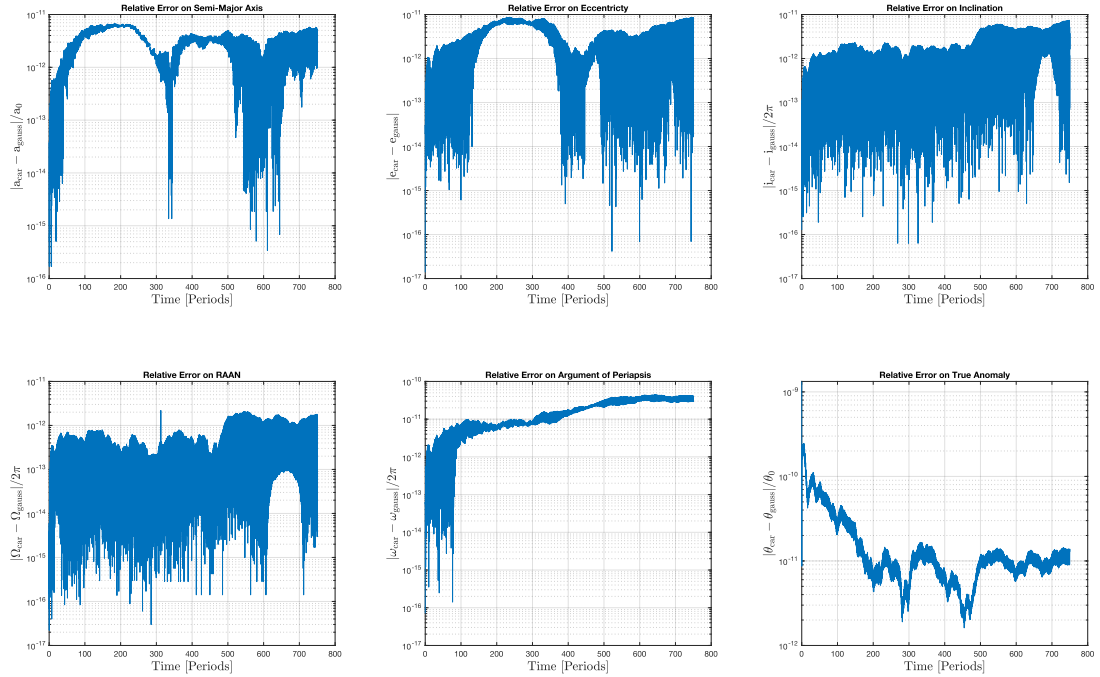


Figure 12: Relative error.

For the nominal propagation time of 750 periods, the short-period oscillations have been filtered in Figure 13. By this way, long period evolutions caused by the SRP have been extracted for the eccentricity, the inclination, the RAAN and the argument of periapsis. On the other hand, secular evolutions for the semimajor axis and the true anomaly are shown. This confirms that the assigned perturbations do not have a secular effect on the semimajor axis. However, the secular evolution of the rest of the elements is not clearly appreciated, especially for the inclination and argument of periapsis. For this reason, a bigger time window needs to be considered and a second propagation for 30,000 periods has been computed.

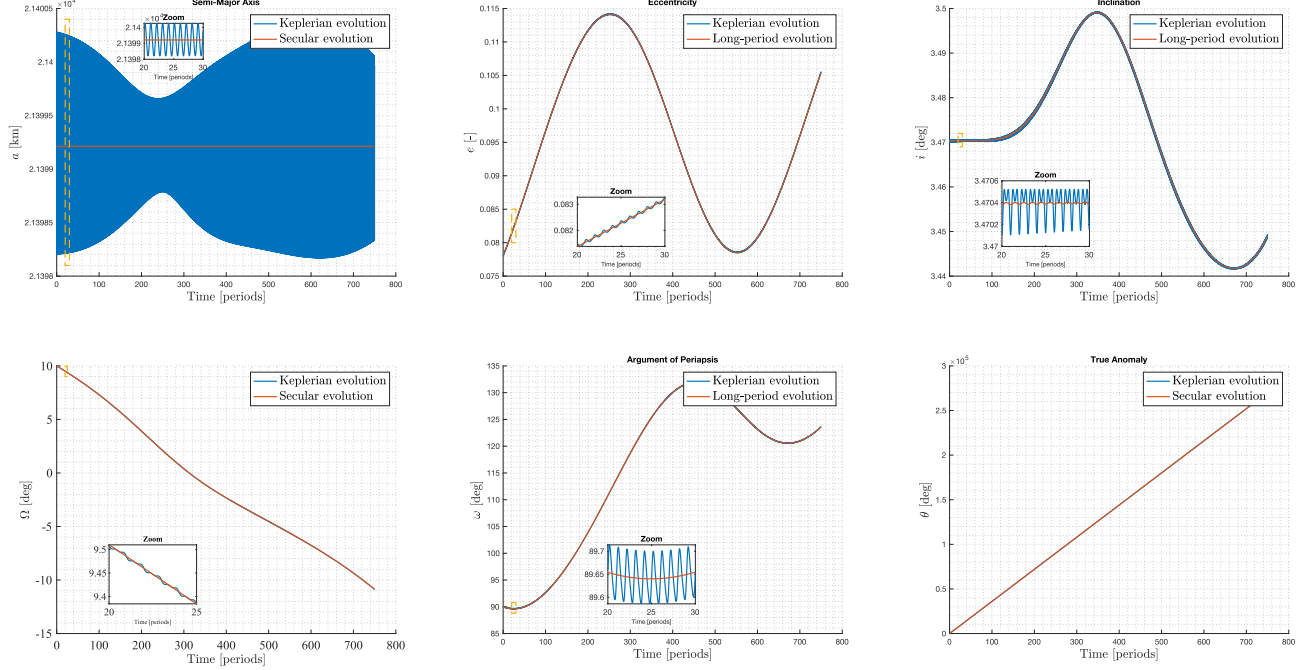


Figure 13: Filtering of short-period oscillations.

In the second propagation, long-period oscillations are filtered. As seen in Figure 14, some interesting longer-period variations appear for the semimajor axis, the eccentricity and the inclination as a second periodic effect of J_2 , and it can also be confirmed that none of them have secular effects. Regarding the RA-AN and the argument of periapsis, the filtering of the longer-period oscillations isolates the secular effects commented in Section 2.4.

2.6 Real data analysis

In this Section, the effect of perturbations on real objects is studied. For this, two uncontrolled objects are chosen in both LEO and MEO orbits and the evolution of their Keplerian elements is shown. The objects considered are:

- **FENGyun 1C DEB (29762)**: a piece of trackable debris that remains in low Earth orbit, as a result of one of the most significant orbital fragmentation events in history. This particular fragment has an area over mass ratio of approximately $A/m = 0.02 \text{ m}^2/\text{kg}$.
- **NAVSTAR 23**: a former satellite of NAVSTAR GPS launched in 1991, and left as an inactive one in a medium Earth orbit. With a cross-sectional area of about 20 m^2 and 1077 kg , its area over mass ratio approximates $A/m = 0.0186 \text{ m}^2/\text{kg}$.

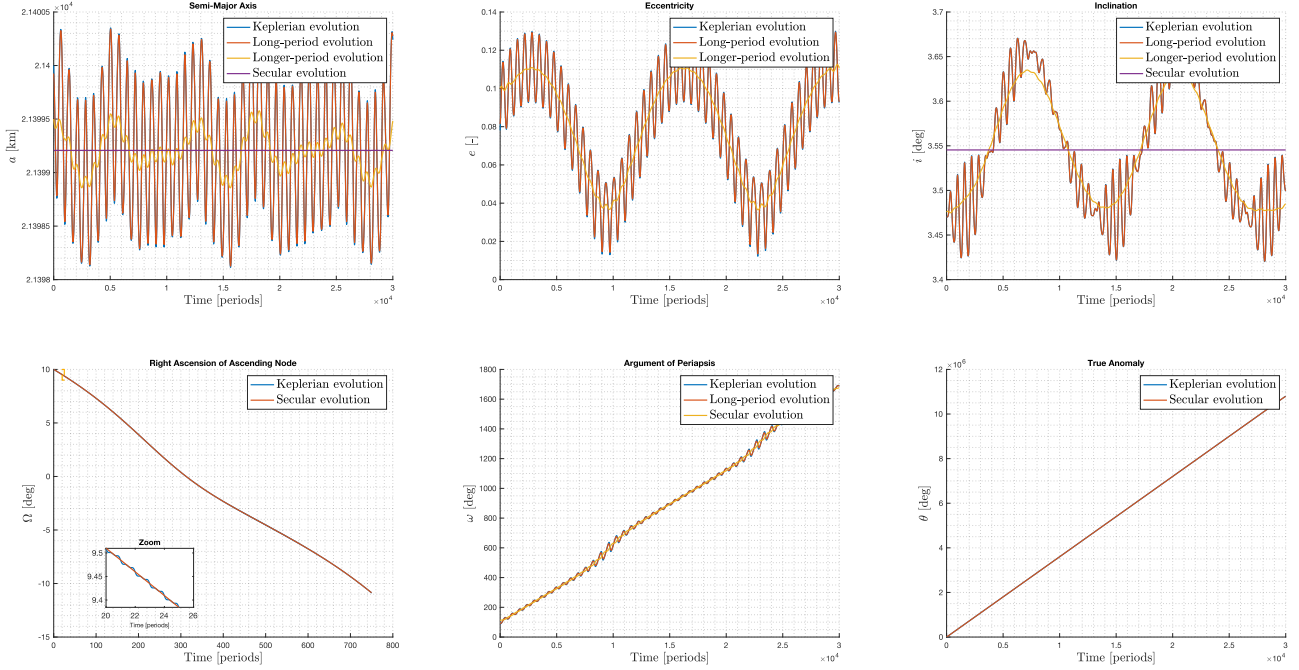


Figure 14: Filtering of long-period oscillations.

To clearly visualize the effects of all the different perturbations, the objects have been propagated using SGP4 for 13 months, from 2023-12-01 to 2024-12-31. To achieve the most accuracy in the results, all the TLEs available at *Space-Track* were provided to the propagator. Furthermore, the evolution of these objects has also been analysed using the model that incorporates J2 and SRP effects, aiming to compare the model's predictions with real-world behaviour.

2.6.1 FENGYUN 1C DEB (29762)

- **Semi-major axis:** the effect of drag can be appreciated in the evolution of the semimajor axis. Due to the importance of this perturbation in LEO orbits, the semimajor axis decreases in the long term. Moreover, short-period oscillations can be noticed due to other perturbations such as J2 or SRP.
- **Eccentricity:** Long period oscillations can be noticed. It could be hypothesized that they are caused by third-body perturbations, but it is more likely that they are caused by the J3 and J4 perturbations, given that their intensity is high in LEO. The influence of SRP in LEO is virtually negligible, this is also why no oscillations with a solar year period are observed in the other orbital elements.
- **Inclination:** short-period oscillations can be pointed out again, as well as a long-period variation. The long-term behaviour can be associated to third body perturbations or triaxiality, although the perturbations from the last one should be small, also J3 and J4 could have a key role on it. It is important not to attribute the long-term behaviour to the effect of SRP, which, as previously mentioned, is negligible in LEO.
- **RA-AN:** it presents short-term oscillations and a positive secular evolution, mainly as a result of J2. The inclination of the orbit induces an eastwards regression of the nodes. Furthermore, from the analysis of the inclination, which tends to be around 98 degrees, it can be observed that the orbit is sun-synchronous. This is reflected in the growth of the RA-AN, which is about 0.986 degrees per day.
- **Argument of perigee:** similarly to the RA-AN, the J2 perturbation induces a negative perigee precession as a secular evolution
- **True anomaly:** No particular behaviour to highlight, except for slight short-term oscillations caused by J2.

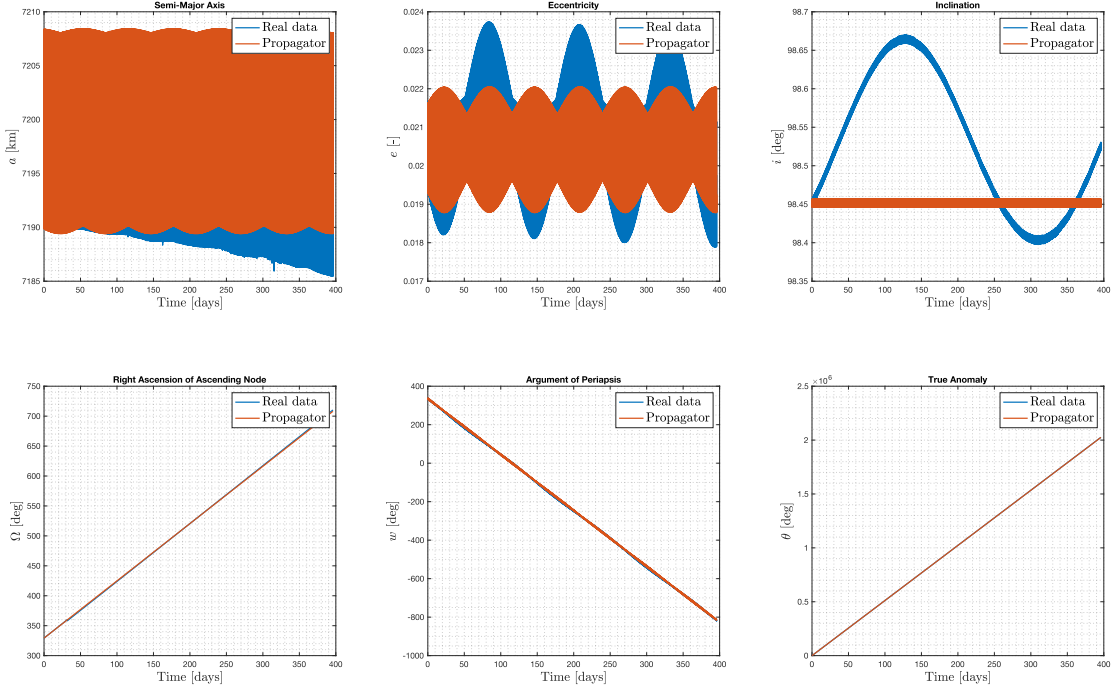


Figure 15: Evolution of Keplerian elements for FENGYUN 1C DEB (29762).

2.6.2 NAVSTAR 23

- **Semi-major axis:** short-period oscillations, mainly due to J2, as can be seen in comparison with the simplified model.
- **Eccentricity:** short-period oscillations, along with a long-period variation due to SRP. In this case, the SRP becomes more relevant as the object is in a higher orbit. This can be proved thanks to its similarity with the simplified model evolution.
- **Inclination:** there seem to be three oscillations with different periods, along with a secular variation. As seen for FENGYUN, there is a big difference between the real data and the model, which is mainly due to third body perturbations.
- **RA-AN:** it can be noticed a westwards node regression due to the effect of J2. This westwards regression can be linked to the inclination value, that keeps being under 90 degrees.
- **Argument of perigee:** short-period oscillations and long-period oscillation due to SRP, consistent with the simplified model. Moreover, there is an positive perigee precession due to secular effect of J2, since the inclination is lower than 63.4 degrees.
- **True anomaly:** similarly to the FENGYUN, short-period oscillations appear due to J2.

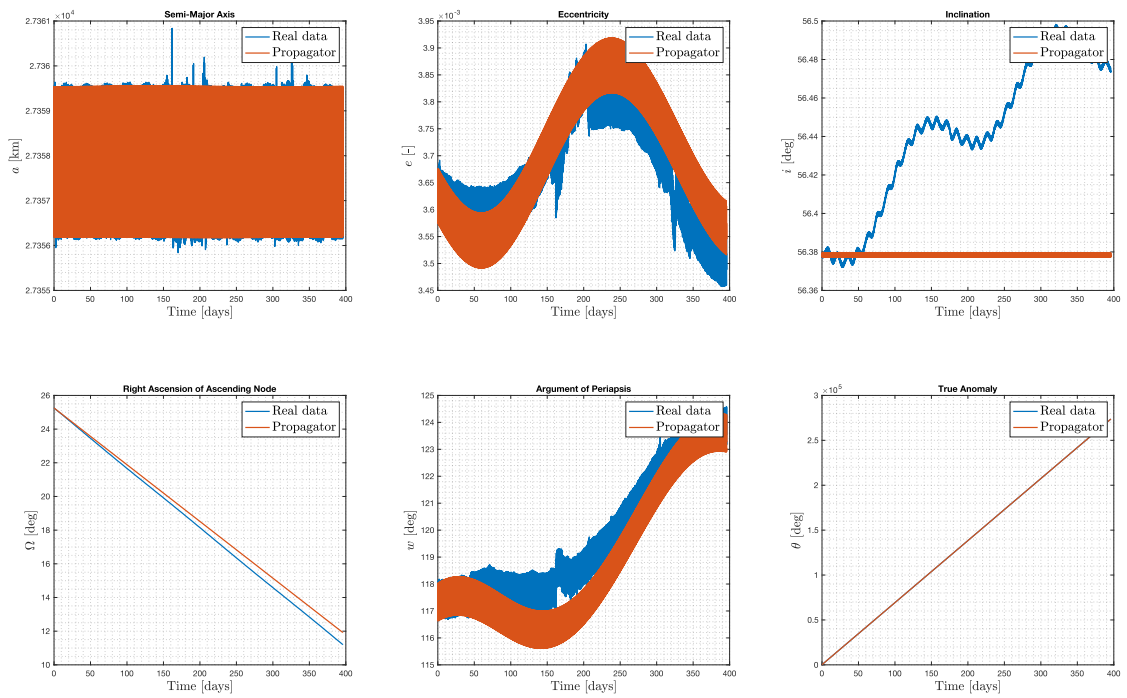


Figure 16: Evolution of Keplerian elements for NAVSTAR 23.